Driving Habits in Older Patients with Central Vision Loss

Sabyasachi Sengupta, DNB, FRCS, Suzanne W. van Landingham, MD, Sharon D. Solomon, MD, Diana V. Do, MD, David S. Friedman, MD, PhD, Pradeep Y. Ramulu, MD, PhD

Objective: To determine if central visual loss is associated with driving cessation, driving restriction, or other-driver preference.

Design: Cross-sectional study.

Participants: Sixty-four subjects with bilateral visual loss (<20/32 in better eye) or severe unilateral visual loss (<20/200) from age-related macular degeneration (AMD) and 58 normally sighted controls between 60 and 80 years of age.

Methods: Participants self-reported driving habits. Other-driver preference was defined as preferring that another drive when there is more than 1 driver in the car. Subjects reporting 2 or more driving limitations were considered to have restricted their driving.

Main Outcome Measures: Self-reported driving cessation, other-driver preference, and driving restriction.

Results: Age-related macular degeneration subjects were older (74.7 vs. 69.7 years), had worse visual acuity (VA; mean better-eye VA, 0.43 vs. 0.08 logarithm of minimum angle of resolution [logMAR]) and contrast sensitivity (CS; 1.4 vs. 1.9 log units of CS [logCS]), and were more likely to be white when compared with controls ($P < 0.001$ for all). Drivers with AMD-related vision loss were more likely to avoid driving over longer distances, beyond 1 hour, at night, and in unfamiliar conditions ($P < 0.05$ for all). In multivariate models, driving cessation was associated with worse better-eye VA (odds ratio [OR], 1.5 per 1-line decrement in VA; $P < 0.001$) and worse binocular CS (OR, 1.36 per 0.1 logCS increment; $P = 0.005$); however, AMD group status was not associated with driving cessation (OR, 1.9; $P = 0.35$). Factors predicting driving restriction were AMD (OR, 9.0; $P = 0.004$), worse vision (OR, 2.5 per line of VA loss; $P < 0.001$), lower CS (OR, 2.2 per 0.1-logCS increment; $P < 0.001$), and female gender (OR, 27.9; $P = 0.002$). Other-driver preference was more common with worse vision (OR, 1.6 per 0.1-logMAR increment; $P = 0.003$), female gender (OR, 4.5; $P = 0.02$), and being married (OR, 3.8; $P = 0.04$).

Conclusions: Most patients with AMD-related central vision loss continue to drive, but demonstrate significant driving restrictions, especially with more severe VA and CS loss. Future work should determine which driving adaptations the visually impaired best balance safety and independence. *Ophthalmology 2014;121:727-732 © 2014 by the American Academy of Ophthalmology.*

Age-related macular degeneration (AMD) is the most frequent cause of severe, irreversible visual impairment in the developed world.[1,2] In the United States alone, there are an estimated 1.75 million adults with neovascular AMD or geographic atrophy, and these cases account for approximately 46% of cases of severe visual loss.[3,4] With improved life expectancies and an aging population, the global prevalence of AMD will rise further in the years to come. Age-related macular degeneration predominantly leads to loss of central vision and is rarely associated with systemic disease (unlike other common diseases such as cataract or diabetes). Additionally, subjects with AMD have well-preserved peripheral vision. Hence, AMD is a good model disease for assessing the effects of central vision loss on important patient outcomes. Prevention of AMD-related vision loss and proper rehabilitation and counseling of individuals with untreatable AMD-related vision loss will become increasingly important, and rehabilitation strategies for visually impaired AMD patients will serve as a model to guide the care of other patients with primarily central vision loss.

Several important functional domains are affected in visually impaired AMD patients, including mobility,[5] reading, recognizing faces, watching TV, and driving.[6] Of these, driving has a particularly strong association with quality of life, especially in the United States, where public transportation is frequently limited or not available. Driving offers independence to older individuals, and limiting or ceasing driving is reported to be associated with incident depression, increased dependence, reduced accessibility to health care, and higher mortality.[7–9] Psychometric tools such as the 25-item National Eye Institute Visual Function Questionnaire have a limited number of questions about driving that have been used to characterize driving impairment on quality of life in patients with AMD[10,11] and have been used to demonstrate that AMD treatments can prevent driving cessation and improve perceived driving ability.[12] However, the small number of driving-related questions on the 25-item National Eye Institute Visual Function Questionnaire precludes a comprehensive description of the types of driving alterations

© 2014 by the American Academy of Ophthalmology
Published by Elsevier Inc.

ISSN 0161-6420/14/$ - see front matter
http://dx.doi.org/10.1016/j.ophtha.2013.09.042

that occur in conjunction with AMD-related vision loss. In particular, there is very little research describing how, and how often, individuals with vision loss resulting from AMD restrict their driving,[13] and literature from other eye diseases suggests that restriction of driving in particular settings may be a more frequent adaptation to vision loss than driving cessation.[14]

In this study, we used a detailed driving questionnaire to compare the driving habits and driving adaptations found in a sample of AMD patients with vision loss and in a sample of normally sighted controls. Evaluated adaptations to driving included complete cessation of driving, restriction of driving to more favorable conditions or locations, and preferring that another person drive, each of which has its own psychosocial impact.[9]

Methods

Study Participants

The study was approved by The Johns Hopkins Medical Institutions' Institutional Review Board, and written informed consent was obtained from all study subjects. Participants were recruited from a convenience sample of patients being cared for at the Wilmer Eye Institute at Johns Hopkins. Subjects' charts were prescreened for eligibility, and eligible subjects had to be between the ages of 60 and 80 years and had to be former or current drivers. Patients were recruited between July 2009 and June 2012.

The AMD subjects were required to have bilateral AMD with evidence of drusen, geographic atrophy, or choroidal neovascularization in both eyes. Visual acuity (VA) was required to be 20/32 or worse in both eyes or worse than 20/200 in one eye regardless of vision in the second eye. Subjects with deterioration of visual angle by more than 50% over the past 3 months and those who received intravitreal injections over the past month were excluded. Control subjects were glaucoma suspects or ocular hypertensives with a VA of 20/40 or better and normal visual fields and were being treated at the same eye hospital as the AMD subjects.[15] Participants also had to be residing in the Mid Atlantic region (defined for this Baltimore-based study as Maryland, Virginia, Delaware, the District of Columbia, and Pennsylvania). Glaucoma suspects were chosen as a comparison group because: (1) they are most likely to have similar social and health behavior characteristics as the AMD patients seeking care at the same hospital and (2) they have been shown to be similar with regard to VA, contrast sensitivity (CS), and visual fields to older individuals without eye disease characterized in a previous population-based survey.[15]

Evaluation of Driving Habits

Driving habits were evaluated with an interviewer-administered questionnaire taken from the Salisbury Eye Evaluation Driving Study, which extends on questionnaires previously used in the Salisbury Eye Evaluation.[16,17] Subjects were asked, "Have you driven a car in the past 3 months?" to assess driving cessation. For those who were currently driving, 9 different driving limitations were assessed: (1) not driving outside of the mid-Atlantic region, (2) not driving more than 1 hour away from home, (3) not driving to neighboring towns or areas, (4) not driving beyond the neighborhood, (5) not driving in the rain, (6) not driving at night, (7) not driving in unfamiliar areas, (8) driving less than twice per week, and (9) driving fewer than 5000 miles per year (the standard for restricted driving in Maryland). Each of these limitations was assessed for the past year, except for driving at night and driving in unfamiliar areas, which were assessed for the past 3 months. The questionnaire asked whether a person had performed a particular driving activity *at all* during the appointed time frame, but not if they chose to avoid that particular driving activity or if they were legally prohibited from doing it.

Driver preference was assessed by asking subjects, "In a typical week when you travel in a car, how often are you the driver?" Subjects who reported that they were the driver 50% or less of the time they rode in a car were considered to have other-driver preference.

Measurement of Vision and Covariates

Monocular VA was measured using the Early Treatment Diabetic Retinopathy Study chart, and better-eye VA was converted to the logarithm of the minimum angle of resolution (logMAR) units for use in statistical analysis. Binocular CS was measured using the Pelli-Robson chart at 1 m with subjects wearing their usual correction and was converted into log units of CS (logCS) for analysis. Both eyes were examined after pupillary dilation for significant lenticular changes in phakic eyes using the Wilmer cataract grading system or posterior capsular opacification in pseudophakic eyes, as previously described.[18,19]

Demographic information was collected by self-report, including age, gender, race, employment status, years of education completed, living situation (if the subject lives with any other adults), and marital status. Cognitive ability was assessed using the Mini-Mental Status Examination for the visually impaired.[20] Depressive symptoms were detected using the Geriatric Depression Scale Short Form, with subjects demonstrating 6 or more positive responses considered to have depressive symptoms.[21] Medical comorbidities were assessed using a standardized structured medical history questionnaire and were summarized as the number of comorbid conditions present.[22] Specific comorbidities inquired about included arthritis, broken or fractured hip, back problems, heart attack or myocardial infarction, angina or chest pain, congestive heart failure, peripheral vascular disease, hypertension, diabetes, emphysema, asthma, stroke, Parkinson's disease, cancer (other than skin cancer), and vertigo or Ménière's disease.

Statistical Analysis

The current study was a secondary analysis from a study powered to detect differences in physical activity and travel patterns in subjects with normal vision, visual field loss resulting from glaucoma, and decreased VA from AMD. The proposed sample size of 60 normally sighted subjects and 60 visually impaired subjects with AMD had a more than 70% power to detect a 3-fold higher odds of driving cessation or driving restriction in our AMD patients as compared with controls assuming a type I error probability of 0.05 and a baseline level of driving cessation or driving restriction similar to that observed in prior population-based surveys.[15]

Group differences for continuous variables were evaluated using the Student t test or Wilcoxon rank-sum test. Chi-square analysis was used to assess differences in categorical variables. Binary outcomes, including driving cessation, driver preference, and the presence of individual driving limitations, were assessed using univariate and multivariate logistic regression models. The number of driving limitations was analyzed as a continuous variable and also was converted into a binary variable, driving restriction, and was classified as present if 2 or more driving limitations were reported.[15] Covariates included in multivariate regression models included those with a P value of less than

0.05 on univariate analysis and those demonstrated to have association with driving habits in previous research.[15] The statistical analysis was completed using Stata software version 12.0 (Stata Corp., College Station, TX).

Results

Sixty-four patients with bilateral AMD and 58 controls were enrolled in the study. Baseline demographics and visual parameters in the 2 groups are shown in Table 1. The AMD patients were older, had worse VA and CS, and were more likely to be white when compared with controls ($P \leq 0.01$ for all). In the better-seeing eye, 30 subjects (47%) with AMD had geographic atrophy and 34 (53%) had a predominantly scarred neovascular membrane involving the center of the fovea.

In univariate analyses, more patients in the AMD group had ceased driving as compared with controls (25.4% vs. 6.9%; $P = 0.006$). Subjects who had stopped driving had significantly worse vision in the better-seeing eye (VA, 0.77 vs. 0.08 logMAR; $P = 0.001$) and CS (logCS, 1.8 vs. 1.2; $P = 0.03$) compared with those who continued to drive. Drivers in the AMD group also had a greater mean number of driving limitations than control group drivers (2.7 vs. 1.1; $P < 0.001$) and were more likely to report 2 or more driving restrictions (Fig 1). Other-driver preference also was reported more frequently by AMD subjects than controls (37% vs. 18%; $P = 0.04$). Presence of cataract or posterior capsular opacification in the better-seeing eye did not show any significant association with driving cessation, restriction, or other-driver preference ($P > 0.5$ for all).

In multivariate models adjusting for age, race, gender, employment, marital status, living situation, cognition, comorbidities, and depressive symptoms, AMD group status was not associated significantly with driving cessation (odds ratio [OR], 1.9; 95% confidence interval [CI], 0.5−7.3; $P = 0.35$; Table 2). However, driving cessation was more likely with worse VA in the better-seeing eye (OR, 1.5 for each 1-line loss of vision; 95% CI, 1.2−1.9; $P < 0.001$) and worse CS (OR, 1.36 for each 0.1-decrement in logCS; 95% CI, 1.1−1.7; $P = 0.005$). Figure 2 shows the modeled probability of driving cessation based on VA among the full study sample. The type of AMD in the better-seeing eye (neovascular vs. nonneovascular) was not associated with driving cessation, nor were any health-related or demographic factors.

More driving restrictions were found among subjects with worse VA (Fig 3). Factors predictive of significant restriction (≥2 driving limitations) of driving included female gender (OR, 27.9; 95% CI, 3.5, 223.0; $P = 0.002$), AMD group status (OR, 9.0; 95% CI, 2.0−40.4; $P = 0.004$), lower CS (OR, 2.2 for each 0.1-decrement in logCS; 95% CI, 1.4−3.5; $P < 0.001$), and worse VA (OR, 2.5 per 0.1-logMAR worsening; 95% CI, 1.6−4.0; $P < 0.001$). Other-driver preference was significantly predicted by worse vision (OR, 1.6 per 0.1-logMAR increment; 95% CI, 1.2−2.1; $P = 0.003$), female gender (OR, 4.5; 95% CI, 1.2−16.2; $P = 0.02$), and being married (OR, 3.8; 95% CI, 1.0−14.3; $P = 0.04$) in multivariate models, whereas lower CS and the AMD group status did not significantly influence other-driver preference ($P > 0.05$ for both).

In multivariate models including both VA and CS, only VA remained associated with driving cessation (OR, 1.4; 95% CI, 1.1−1.9; $P = 0.01$) and driving restriction (OR, 2.1; 95% CI, 1.2−3.4; $P = 0.007$), although these 2 visual measures were correlated significantly ($r = 0.78$; $P < 0.001$). The association between VA and both driving cessation and restriction also persisted when the presence or absence of cataract or posterior capsular opacification was included in the models.

Table 1. Characteristics of Study Participants in Age-Related Macular Degeneration Group versus Controls

Parameter Evaluated	Controls (n = 58)	Age-Related Macular Degeneration (n = 64)	P Value
Demographics			
Age (yrs)	69.8 (5.29)	74.8 (4.98)	<0.001
Female gender (%)	60.3	57.1	0.72
White race (%)	77.6	93.7	0.01
Education (years)	15.5 (2.2)	15 (1.9)	0.13
Unemployment (%)	60.3	79.3	0.02
Lives with others (%)	25.9	17.5	0.26
Married (%)	67.2	68.3	0.11
Health/cognition			
MMSE-VI score	20.8 (1.4)	20.6 (1.7)	0.76
Comorbid illnesses (no.)	2 (2, 3)	2.5 (1, 3)	0.54
Depressive symptoms (%)	5.2	4.8	0.9
Vision parameters			
Binocular CS (log units)	1.9 (0.12)	1.4 (0.33)	<0.001
Better eye acuity (logMAR)	0.08 (0.12)	0.43 (0.32)	<0.001
Worse eye acuity (logMAR)	0.18 (0.14)	1.05 (0.46)	<0.001
Significant cataract/PCO			
Right eye (%)	16.7	27.6	0.16
Left eye (%)	14.8	23.3	0.25

CS = contrast sensitivity; logMAR = logarithm of the minimum angle of resolution; MMSE-VI = Mini-Mental Status Examination; PCO = posterior capsular opacification.
Continuous variables reported as mean (95% confidence interval).

Discussion

One in 4 patients with AMD-related central vision loss was noted to have stopped driving in our cohort, although patients in this group were not more likely than controls to have stopped driving in multivariate models. Driving cessation was, however, more common with worse VA in the better-seeing eye and CS, suggesting that driving cessation is more likely in more advanced disease. Although most of our visually impaired AMD patients continued to

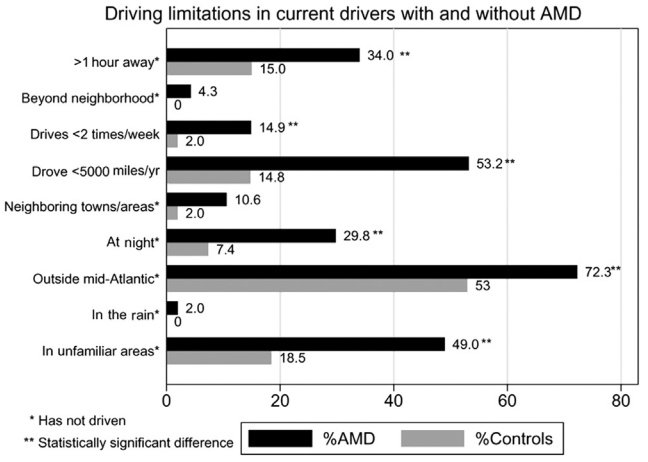


Figure 1. Bar graph showing the driving limitations in current drivers with and without age-related macular degeneration (AMD).

Table 2. Multivariate Analysis Showing Effect of Age-Related Macular Degeneration and Its Severity on Driving Status

Variable	Interval	Not Driving	Driving Restriction*	Prefer Another Driver*
Vision				
AMD	Present	1.9 (0.5–7.3)	9.0 (2.0–40.4)§	2.2 (0.7–7.1)
Type of AMD (worse eye)	Wet AMD	0.6 (0.1–3.3)	0.4 (0.06–1.9)	0.3 (1.8)
Type of AMD (better eye)	Wet AMD	2.7 (0.6–11.5)	0.7 (0.2–2.9)	1.2 (0.3–4.8)
Binocular contrast sensitivity	1 letter worse†	1.36 (1.1–1.7)§	2.2 (1.4–3.5)§	1.24 (0.9–1.6)
Better eye acuity (logMAR)	1 line worse‡	1.5 (1.2–1.9)§	2.5 (1.6–4.0)§	1.6 (1.1–2.1)§
Demographics				
Age	5 yrs older	1.9 (0.8–4.4)	1.8 (0.8–4.2)	1.0 (0.6–1.8)
Race	White	—	0.3 (0.03–1.8)	1.4 (0.3–7.0)
Gender	Female	1.1 (0.3–4.3)	27.9 (3.5–224)§	4.5 (1.2–16.2)§
Employment	Present	0.2 (0.02–2.2)	0.3 (0.05–1.9)	0.3 (0.06–1.2)
Living situation	Lives with others	0.4 (0.06–3.1)	3.2 (0.6–17.1)	2.0 (0.6–7.2)
Marital status	Married	1.2 (0.2–5.9)	5.4 (0.9–31.2)	3.8 (1.0–14.2)§
Health/cognition				
MMSE VI score	5 points lower	1.1 (0.2–7.3)	1.4 (0.2–10.5)	0.7 (0.1–4.1)
Comorbidities	1 illness	1.1 (0.7–1.6)	0.8 (0.5–1.3)	1.1 (0.7–1.6)
Depressive symptoms	Present	2.1 (0.1–35.3)	0.06 (0.001–3.3)	—

AMD = age-related macular degeneration; logMAR = logarithm of minimal angle of resolution; MMSE-VI = Mini-Mental Status Examination.
Data are odds ratio (95% confidence interval).
*Refers to current drivers only.
†Corresponds to 0.05-log unit change.
‡Corresponds to 0.1-logMAR change.
§$P < 0.05$.

drive, they were significantly more likely to prefer that another adult drive when possible and were more likely to restrict where, when, and how far they drive. These findings suggest that driving restriction, and not driving cessation, is the primary driving adaptation found among patients with AMD-associated vision loss.

Our findings regarding driving cessation differ somewhat from previous papers examining driving cessation in AMD. Campbell et al[22] found that approximately 50% of their patients with AMD had ceased driving and that AMD was

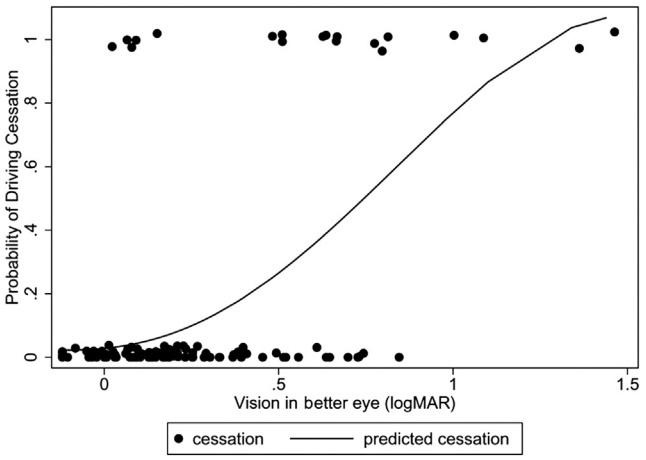


Figure 2. Graph showing the modeled probability of not driving as a function of better-eye visual acuity in age-related macular degeneration patients. In addition to better-eye visual acuity, our multivariate logistic regression model includes age, gender, unemployment, cognition, comorbidities, and depressive symptoms. logMAR = logarithm of minimum angle of resolution.

1 of the 6 most important causes of driving cessation in the elderly. However, neither severity of AMD in terms of vision loss nor type of AMD was recorded in this study. DeCarlo et al[24] used the Driving Habits Questionnaire[23] to characterize AMD patients seeking low-vision rehabilitation and reported that only 24% patients in their series continued driving. The lower frequency of driving cessation noted in the current study suggests that driving cessation is much more dependent on severity of vision loss in the better-seeing eye rather than the mere presence or absence of AMD or AMD subtype. Indeed, many of our studied patients met the state of Maryland's requirements for an unrestricted (at least 20/40 in 1 eye) or restricted (20/70 in at least 1 eye) driver's license, allowing them the opportunity to continue driving if they were so inclined.

Many patients with central vision loss may adjust to driving difficulty by self-regulating when, where, and how often they drive as opposed to stopping driving altogether. Of note, the apparent preference to restrict driving as a result of AMD-related vision loss, as opposed to stopping driving altogether, is distinct from what is observed with visual field loss in glaucoma, where driving cessation is an adaptation as much or more frequently seen than driving restriction.[14,15] Our group of visually impaired AMD patients reported increased driving limitations, with almost 1 in 3 patients avoiding driving at night, 1 in 2 avoiding driving in unfamiliar areas, and 3 in 4 avoiding driving over long distances. DeCarlo et al[24] found significantly reduced driving exposure and frequent avoidance of challenging driving situations among advanced AMD patients requiring low-vision aids who continued to drive, although no control group was available for comparison. Ball et al,[25] using a driving habits questionnaire, reported more driving avoidance in subjects

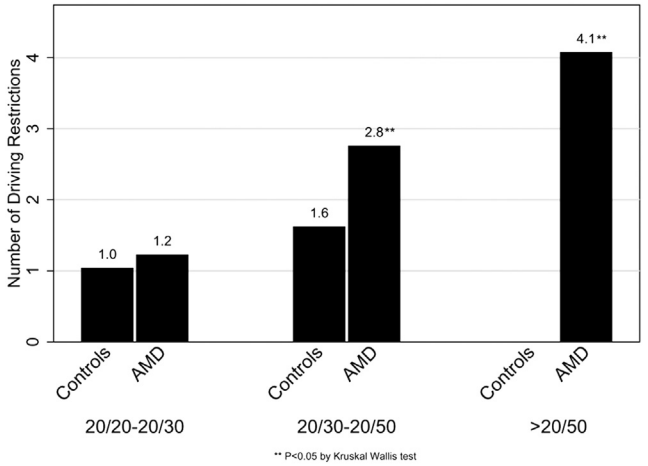


Figure 3. Bar graph showing the mean number of driving restrictions in age-related macular degeneration (AMD) and control subjects.

with AMD and cataract, although AMD subjects comprised less than 10% (n = 19) of the total sample studied. Another interview-based study conducted by Moore and Miller[26] found that the older drivers with AMD used several strategies while driving, such as exercising extreme caution, relying on memory for the location of turn-offs and traffic control devices, and increased scanning of the road, suggesting that driving is significantly more challenging in visually impaired AMD patients who continue to drive.

We found loss of VA in the better-seeing eye to be the best predictor for driving cessation, restriction, and other-driver preference. In a recent study of pooled data from the MARINA (Minimally Classic/Occult Trial of the Anti-VEGF Antibody Ranibizumab in the Treatment of Neovascular Age-Related Macular Degeneration trial) and ANCHOR (Anti-VEGF Antibody for the Treatment of Predominantly Classic Choroidal Neovascularization in Age-Related Macular Degeneration) trials, Bressler et al[12] reported that patients receiving anti−vascular endothelial growth factor therapy for choroidal neovascularization were less likely to stop driving and had better perception of driving ability as compared with subjects receiving sham treatments. In these studies, more than 80% patients with AMD continued to drive at the 2-year follow-up period, similar to our data, although the frequency of driving limitation was not explored. Together, these data strongly suggest that preservation of VA may enable elderly subjects with AMD to sustain and enjoy their driving capabilities.

In addition to worse VA in the better-seeing eye, we found that reduction in CS was associated significantly with driving cessation and restriction, but not driving preference, similar to results reported from glaucoma patients.[14] Bronstad et al,[27] in a recent study using driving simulators, showed that CS was the best predictor for responses when the participants with AMD perceived virtual pedestrians using the areas of the visual field with no scotoma, similar to our results. In this study, VA was not correlated with response measures, that is, late and missed responses. In our study, when both VA and CS were included in predictive models for driving cessation and restriction, we found that only VA was associated significantly with driving cessation and restriction, although both VA and CS were highly correlated. Our current study also found that women were considerably more likely to have driving restriction and that married individuals were more likely to prefer another to drive as compared with controls. Campbell et al[22] also showed similar trends and in their study, with women twice as likely as men to stop driving (although driving restriction and other-driver preference were not measured by the authors). This trend suggests that women may be inherently more cautious and hence restrict themselves more often than men, and also that married women probably frequently defer driving to their spouse.

One drawback of our study is the use of self-reported driving habits from patients, which may differ in different settings. Additionally, we used glaucoma suspects as controls in our study and not so-called truly normal subjects, although previous work has demonstrated that the visual characteristics of this group are almost identical to those of normal subjects from population-based surveys.[15] In some aspects, this control group is better suited for comparison with our AMD cases because they match the AMD population with respect to seeking care at a tertiary referral care center, which may reflect very specific behavioral patterns and may have very low rates of driving cessation and driving restriction (suggesting that they are indeed visually normal despite their suspect glaucoma). Although we realize that intravitreal injections are a routine part of care for wet AMD, we excluded subjects who received intravitreal injections within 1 month of taking the questionnaire to minimize the effect of recent visual change. Given emerging evidence for the need for long-term maintenance of anti−vascular endothelial growth factor therapy in neovascular AMD,[28] future studies would do well to also study patients receiving anti−vascular endothelial growth factor therapy. Finally, because our study was cross-sectional, we cannot determine when, or at what level of VA, our AMD patients stopped or restricted their driving. A strength of our study was the use of a detailed questionnaire to determine various aspects of driving. The use of this questionnaire allowed us to document the many ways that visually impaired AMD patients restrict their driving as compared with controls including distance, frequency, conditions, and driver preference.

In conclusion, most of the visually impaired AMD patients in our series continued to drive, but had significant driving restrictions. Further work is necessary to evaluate whether these restrictions are effective adaptations that effectively balance safety and independence, or whether they simply indicate that patients minimize their total driving risk by driving in dangerous conditions less often.

References

1. Klein R, Klein BE, Linton KL. Prevalence of age-related maculopathy. The Beaver Dam Eye Study. Ophthalmology 1992;99:933–43.
2. Kahn HA, Leibowitz HM, Ganley JP, et al. The Framingham Eye Study. I. Outline and major prevalence findings. Am J Epidemiol 1977;106:17–32.

3. Eye Diseases Prevalence Research Group. Prevalence of age-related macular degeneration in the United States. Arch Ophthalmol 2004;122:564–72.
4. Eye Diseases Prevalence Research Group. Causes and prevalence of visual impairment among adults in the United States. Arch Ophthalmol 2004;122:477–85.
5. Popescu ML, Boisjoly H, Schmaltz H, et al. Age-related eye disease and mobility limitations in older adults. Invest Ophthalmol Vis Sci 2011;52:7168–74.
6. Mitchell J, Bradley C. Quality of life in age-related macular degeneration: a review of the literature. Health Qual Life Outcomes [serial online] 2006;4:97. Available at: http://www.hqlo.com/content/4/1/97. Accessed August 11, 2013.
7. Fonda SJ, Wallace RB, Herzog AR. Changes in driving patterns and worsening depressive symptoms among older adults. J Gerontol B Psychol Sci Soc Sci 2001;56:S343–51.
8. Edwards JD, Perkins M, Ross LA, Reynolds SL. Driving status and three-year mortality among community-dwelling older adults. J Gerontol A Biol Sci Med Sci 2009;64:300–5.
9. Freeman EE, Gange SJ, Muñoz B, West SK. Driving status and risk of entry into long-term care in older adults. Am J Public Health 2006;96:1254–9.
10. Hochberg C, Maul E, Chan ES, et al. Association of vision loss in glaucoma and age-related macular degeneration with IADL disability. Invest Ophthalmol Vis Sci 2012;53:3201–6.
11. Finger RP, Fleckenstein M, Holz FG, Scholl HP. Quality of life in age-related macular degeneration: a review of available vision-specific psychometric tools. Qual Life Res 2008;17:559–74.
12. Bressler NM, Chang TS, Varma R, et al. Driving ability reported by neovascular age-related macular degeneration patients after treatment with ranibizumab. Ophthalmology 2013;120:160–8.
13. Owsley C, McGwin G Jr. Driving and age-related macular degeneration. J Vis Impair Blind 2008;102:621–35.
14. van Landingham SW, Hochberg C, Massof RW, et al. Driving patterns in older adults with glaucoma. BMC Ophthalmol [serial online] 2013;13:4. Available at: http://www.biomedcentral.com/14712-415/13/4. Accessed August 11, 2013.
15. Ramulu PY, West SK, Munoz B, et al. Driving cessation and driving limitation in glaucoma: the Salisbury Eye Evaluation Project. Ophthalmology 2009;116:1846–53.
16. Freeman EE, Muñoz B, Turano KA, West SK. Measures of visual function and their association with driving modification in older adults. Invest Ophthalmol Vis Sci 2006;47:514–20.
17. Turano KA, Munoz B, Hassan SE, et al. Poor sense of direction is associated with constricted driving space in older drivers. J Gerontol B Psychol Sci Soc Sci 2009;64:348–55.
18. West SK, Munoz B, Wang F, Taylor H. Measuring progression of lens opacities for longitudinal studies. Curr Eye Res 1993;12:123–32.
19. Findl O, Buehl W, Menapace R, et al. Comparison of 4 methods for quantifying posterior capsule opacification. J Cataract Refract Surg 2003;29:106–11.
20. Busse A, Sonntag A, Bischkopf J, et al. Adaptation of dementia screening for vision-impaired older persons: administration of the Mini-Mental State Examination (MMSE). J Clin Epidemiol 2002;55:909–15.
21. Montorio I, Izal M. The Geriatric Depression Scale: a review of its development and utility. Int Psychogeriatr 1996;8:103–12.
22. Campbell MK, Bush TL, Hale WE. Medical conditions associated with driving cessation in community dwelling, ambulatory elders. J Gerontol 1993;48:S230–4.
23. Owsley C, Stalvey B, Wells J, Sloane ME. Older drivers and cataract: driving habits and crash risk. J Gerontol A Biol Sci Med Sci 1999;54:M203–11.
24. DeCarlo DK, Scilley K, Wells J, Owsley C. Driving habits and health-related quality of life in patients with age-related maculopathy. Optom Vis Sci 2003;80:207–13.
25. Ball K, Owsley C, Stalvey B, et al. Driving avoidance and functional impairment in older drivers. Accid Anal Prevent 1998;30:313–22.
26. Moore L, Miller M. Driving strategies used by older adults with macular degeneration: assessing the risks. Appl Nurs Res 2005;18:110–6.
27. Bronstad PM, Bowers AR, Albu A, et al. Driving with central field loss. I. Effect of central scotomas on responses to hazards. JAMA Ophthalmol 2013;131:303–9.
28. Rofagha S, Bhisitkul RB, Boyer DS, et al; SEVEN-UP Study Group. Seven-year outcomes in ranibizumab-treated patients in ANCHOR, MARINA, and HORIZON: a multicenter cohort study (SEVEN-UP). Ophthalmology 2013;120:2292–9.

Footnotes and Financial Disclosures

Originally received: April 16, 2013.
Final revision: September 23, 2013.
Accepted: September 25, 2013.
Available online: December 2, 2013. Manuscript no. 2013-623.

Wilmer Eye Institute, The Johns Hopkins University, Baltimore, Maryland.

Financial Disclosure(s):
The author(s) have no proprietary or commercial interest in any materials discussed in this article.

Supported in part by the Dennis W. Jahnigen Memorial Award (American Geriatrics Society, New York, New York); the National Institutes of Health, Bethesda, Maryland (grant no.: EY018595); Research to Prevent Blindness Robert and Helen Schaub Special Scholar Award, New York, New York; the Intramural Research Program of the National Institute on Aging, National Institutes of Health, Bethesda, Maryland; and the Doris Duke Charitable Research Foundation Clinical Research Fellowship, New York, New York. All funding organizations had no role in the design or conduct of this research.

Correspondence:
Pradeep Y. Ramulu, MD, PhD, Wilmer Eye Institute, The Johns Hopkins University, 600 North Wolfe Street, Maumenee B-110, Baltimore, MD 21287. E-mail: pramulu1@jhmi.edu.